



2020 HSC Course Assessment Task 3

1st of July, 2020

- Working time – 3 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

- Mark your answers on the answer sheet provided (numbered as page 13)

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class (please ✓)

○ 12M4C – Ms Ziaziaris

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions

Marks

1. 1
 Given $z = -1 - i$ and $\omega = 1 + i\sqrt{3}$. The Euler form of $\frac{z}{\omega}$ is
 (A) $\sqrt{2}e^{\frac{11i\pi}{12}}$ (C) $-\frac{1}{\sqrt{2}}e^{\frac{-i\pi}{12}}$
 (B) $-\sqrt{2}e^{\frac{-i\pi}{12}}$ (D) $\frac{1}{\sqrt{2}}e^{\frac{11i\pi}{12}}$
2. 1
 The unit vector in the same direction as $\vec{u} = 4\vec{i} + 4\vec{j} - 2\vec{k}$ is
 (A) $\frac{-1}{36} \begin{bmatrix} -4 \\ -4 \\ 2 \end{bmatrix}$ (C) $\frac{1}{5} \begin{bmatrix} 4 \\ 4 \\ -2 \end{bmatrix}$
 (B) $\frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$ (D) $\frac{1}{6} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$
3. 1
 Which of the following is an expression for $\int \frac{\cos^3 x - \sin^3 x}{\cos x - \sin x} dx$?
 (A) $x + \frac{1}{2} \cos 2x + c$ (C) $x + \frac{1}{2} \sin^2 x + c$
 (B) $x - \frac{1}{2} \cos 2x + c$ (D) $x - \frac{1}{2} \sin^2 x + c$
4. 1
 A particle of mass m falls vertically from rest under gravity in a medium in which the resistance to motion has magnitude $\frac{1}{40}mv^2$ where $v \text{ ms}^{-1}$ is the speed of the particle and $g = 9.8 \text{ ms}^{-2}$ is the acceleration due to gravity. What is the terminal velocity of the particle?
 (A) 19.8 ms^{-1} (C) 9.8 ms^{-1}
 (B) 28 ms^{-1} (D) 40 ms^{-1}

5. If ω is a complex cube root of unity and $\omega \neq 1$. What is the value of 1

$$\left(1 + \frac{1}{\omega}\right)^{2020} \left(1 + \frac{1}{\omega^2}\right)^{2021}.$$

(A) $-\frac{1}{\omega}$ (B) ω (C) 0 (D) 1

6. The projection of $\overrightarrow{OA}$ onto $\overrightarrow{OB}$ for $A(4, 2, -3)$ and $B(-1, 1, 1)$. 1

(A) $\frac{5}{3} \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$ (C) $\frac{5}{3} \begin{bmatrix} 4 \\ 2 \\ -3 \end{bmatrix}$

(B) $\frac{5}{3} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ (D) $-\frac{5}{3} \begin{bmatrix} 4 \\ 2 \\ -3 \end{bmatrix}$

7. If the complex number z satisfies $|z| - z - 4(1 - 2i) = 0$, which of the following is $|z|^2$? 1

(A) 80 (B) 180 (C) 100 (D) 400

8. A particle of mass m is moving horizontally in a straight line. It experiences a resistive force of magnitude $3m(v + v^2) N$ when its speed is v metres per second. At time t seconds, the particle has a displacement of x metres from a fixed point O . Which of the following is the correct expression for x in terms of v ? 1

(A) $x = -\frac{1}{3} \int \frac{1}{1+v} dv$ (C) $x = -\frac{1}{3} \int \frac{1}{v(1+v)} dv$

(B) $x = \frac{1}{3} \int \frac{1}{1+v} dv$ (D) $x = \frac{1}{3} \int \frac{1}{v(1+v)} dv$

9. The value of $\frac{d}{dx} \left(\int_x^{x^2} \frac{1}{t-1} dt \right)$ is 1

(A) $\frac{-x}{x^2-1}$ (B) $\frac{1}{x-1}$ (C) $\frac{1}{x+1}$ (D) $\frac{x}{x+1}$

10. The negation of the following statement: 1

" $\forall p \in P$ (p is of the form $4m + 1 \Rightarrow p$ can be written as a sum of two squares)" is

- (A) $\forall p \in P, p$ is of the form $4m + 1$ and p can not be written as a sum of two squares.
- (B) $\exists p \in P, p$ is not of the form $4m + 1$ and p can be written as a sum of two squares.
- (C) $\forall p \in P, p$ is not of the form $4m + 1$ or p can not be written as a sum of two squares.
- (D) $\exists p \in P, p$ is of the form $4m + 1$ and p can not be written as a sum of two squares.

End of Section I

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Given $z = 2 + 2i\sqrt{3}$ and $\omega = \sqrt{3} + i$		
i. Find $z\omega$ in Cartesian form		1
ii. show that $\frac{z}{\omega} = \omega$		1
(b) Find $\int \frac{x}{\sqrt{1-x}} dx$		2
(c) i. Find $\sqrt{6i-8}$.		2
ii. Hence, solve the equation		2
$2z^2 - (3+i)z + 2 = 0.$		
(d) i. Find a, b and c such that		2
$\frac{16}{(x^2+4)(2-x)} = \frac{ax+b}{x^2+4} + \frac{c}{2-x}.$		
ii. Hence, find $\int \frac{1}{(x^2+4)(2-x)} dx$		2
(e) Given that $\vec{u} = 2\vec{i} + 2\vec{j} - 3\vec{k}$ and $\vec{v} = \vec{i} + 2\vec{j} + 2\vec{k}$		
i. Show that $\vec{u}$ and $\vec{v}$ are perpendicular.		1
ii. Hence, or otherwise find $\vec{w}$ such that		2
$ \vec{w} ^2 = \vec{u} ^2 + \vec{v} ^2.$		

End of Question 11

Question 12 (15 Marks)

Commence a NEW page.

Marks

- (a) Given that
- m
- and
- n
- are integers.

Statement: "If mn is odd, then m and n are odd".

i. Write down the contrapositive of the statement.

1

ii. Proving the statement by proving its contrapositive.

2

- (b) The velocity of a particle moving in a straight line is given by

$$v^2 = -9x^2 + 18x + 27.$$

i. Prove that the motion is simple harmonic.

2

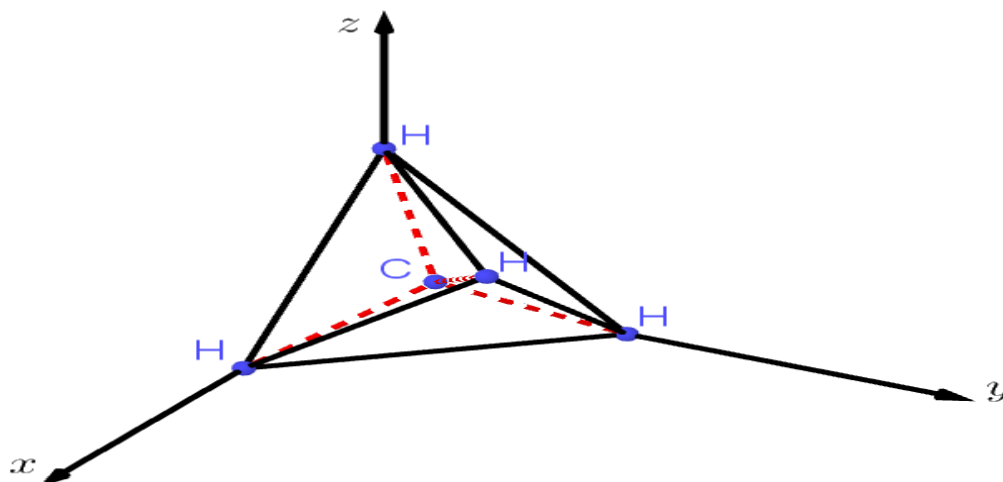
ii. Hence, find the amplitude and the centre of the motion.

2

iii. Find the maximum acceleration of the particle and state where it occurs.

2

- (c) A molecule of methane
- CH_4
- , is structured with the four hydrogen atoms at the vertices of a regular tetrahedron and the carbon atom at the centroid. The bond angle is the angle formed by the
- $H-C-H$
- combination; It is the angle between the lines that join the carbon atom to two of the hydrogen atoms.



Consider the vertices of the tetrahedron to be the points $A(1, 0, 0)$, $B(0, 1, 0)$, $D(0, 0, 1)$ and $E(1, 1, 1)$ where, A, B, D and E representing the hydrogen atoms as shown in the figure. The carbon atom is represented by point C .

- i. Given that

$$\overrightarrow{CA} + \overrightarrow{CB} + \overrightarrow{CD} + \overrightarrow{CE} = \mathbf{0}.$$

2

Show that the centroid is given by $C(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

- ii. Show that the bond angle is about
- 109.5°
- .

2

- iii. Find the equation of the line passing through the point
- C
- and perpendicular to both
- $\overrightarrow{AB}$
- and
- $\overrightarrow{AD}$
- .

2**End of Question 12**

Question 13 (15 Marks)

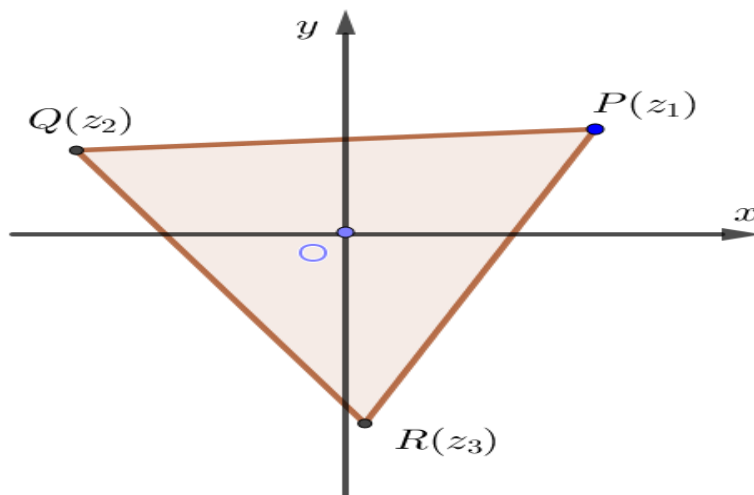
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Marks

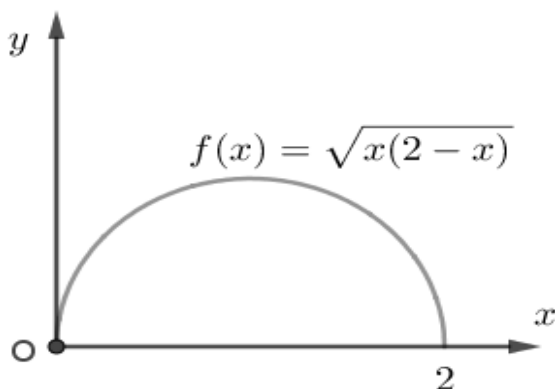
- (a) On an Argand diagram, sketch the region represented by the complex number z where **3**

$$0 \leq \arg(z) \leq \frac{\pi}{4}, \quad |z - 2 - 2i| \geq 2 \quad \text{and} \quad z + \bar{z} < 8.$$

- (b) The point P represents the complex number $z_1 = 3 + 2i$ on the Argand diagram. The complex numbers z_2 and z_3 represented by Q and R respectively, so that PQR is an equilateral triangle whose centre is at the origin. Let $\omega = e^{\frac{i\pi}{3}}$.



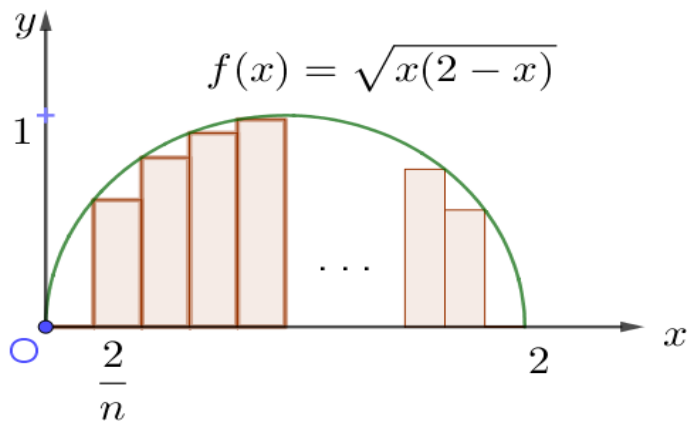
- i. Show that $z_2 = \omega^2 z_1$ and $z_3 = \omega^4 z_1$ **2**
- ii. Find the complex number z_4 represented by the point S such that $PQRS$ is a parallelogram. **2**
- (c) Consider the function $f(x) = \sqrt{x(2-x)}$, see diagram below.



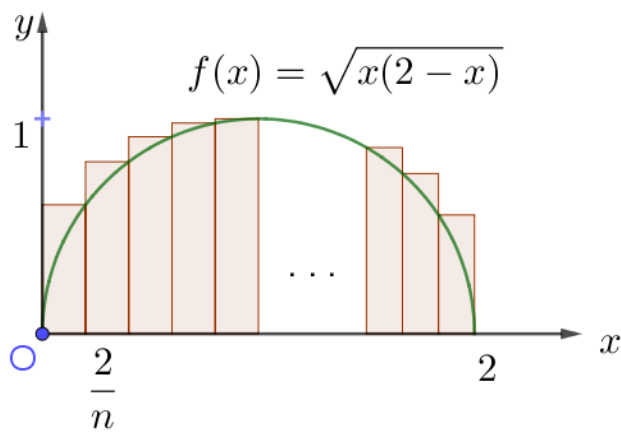
- i. Evaluate $\int_0^2 f(x) dx$. **1**

Question 13 continues

- ii. In the diagram below, the interval $[0, 2]$ is divided into n equal parts. Find the sum of the area of the shaded region. **2**



- iii. Find the sum of the area of the shaded region in the diagram below. **2**



- iv. Hence or otherwise, show that $\lim_{n \rightarrow \infty} \sum_{r=1}^{r=n} \frac{\sqrt{r(n-r)}}{n^2} = \frac{\pi}{8}$ **3**

End of Question 13

Question 14 (15 Marks)

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Marks

- (a) A particle A is dropped from a weather balloon. The equation of motion of particle A is

$$\ddot{x} = g - kv_A$$

where g is the acceleration due to gravity, k is a positive constant and v_A is the velocity of particle A .

T seconds later, an identical particle B is projected downwards from the same weather balloon with initial velocity $u \text{ ms}^{-1}$. The equation of motion of particle B is

$$\ddot{x} = g - kv_B$$

where v_B is the velocity of particle B .

- i. Show that

4

$$T = -\frac{1}{k} \ln \left(\frac{g - ku}{g} \times \frac{g - kv_A}{g - kv_B} \right).$$

- ii. Show that particle B 's displacement x_B is given by

3

$$x_B = \frac{1}{k} \left[u - v_B + \frac{g}{k} \ln \left(\frac{g - ku}{g - kv_B} \right) \right].$$

- iii. Deduce that if particle B catches up with particle A , then particle B must have been released no more than $\frac{u}{g}$ seconds after particle A .

3

- (b) A sequence is defined by $y_0 = 2, y_1 = 2 \cos x - i \sin x$,

$$y_{n+1} = 2 \cos x y_n - y_{n-1}, \quad n \geq 1 \quad \text{and} \quad 0 \leq x \leq \frac{\pi}{2}$$

- i. Use mathematical induction to prove that

3

$$y_n = \frac{1}{2} (\cos x + i \sin x)^n + \frac{3}{2} (\cos x - i \sin x)^n \quad n \geq 0.$$

- ii. Hence or otherwise, show that $\lim_{n \rightarrow \infty} \frac{y_{n+1}}{y_n} = e^{ix}$.

2**End of Question 14**

Question 15 (15 Marks)

Commence a NEW page.

Marks

- (a) Consider ω is an n -th root of unity. You may assume that

$$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0.$$

- i. Hence, or otherwise, show that

2

$$1 + 2\omega + 3\omega^2 + \cdots + n\omega^{n-1} = \frac{n}{\omega - 1}.$$

- ii. By expressing $z^n - 1$ as product of its factors, deduce that

2

$$(1 - \omega)(1 - \omega^2) \cdots (1 - \omega^{n-1}) = n.$$

- iii. Let $P(z) = (z - \omega)(z - \omega^2)(z - \omega^3) \cdots (z - \omega^{n-1})$.

A. Find $\frac{P'(z)}{P(z)}$.

2

- B. Hence, find

2

$$\frac{1}{1 - \omega} + \frac{1}{1 - \omega^2} + \cdots + \frac{1}{1 - \omega^{n-1}}.$$

- (b) Let $x = \frac{a}{a-b}$, $y = \frac{b}{b-c}$ and $z = \frac{c}{c-a}$, where a, b and c are real numbers.

- i. Show that $(x-1)(y-1)(z-1) = xyz$

2

- ii. Hence or otherwise, show that

2

$$x + y + z = xy + yz + zx + 1.$$

- iii. Use (i) and (ii) to show that

3

$$\left(\frac{2a-b}{a-b}\right)^2 + \left(\frac{2b-c}{b-c}\right)^2 + \left(\frac{2c-a}{c-a}\right)^2 \geq 5.$$

End of Question 15

Question 16 (15 Marks)

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Marks

Given that

$$I_n = \int_0^{\frac{\pi}{4}} (1 + \tan x)^n dx$$

i. Show that

4

$$I_n = \frac{2^{n-1} - 1}{n - 1} + 2I_{n-1} - 2I_{n-2}, \quad n \geq 2$$

ii. Hence or otherwise, evaluate I_5 **3**

iii.

Show that

3

$$I_n = \frac{2^{n-1} - 1}{n - 1} + 2 \frac{2^{n-2} - 1}{n - 2} + 2 \frac{2^{n-3} - 1}{n - 3} - 4I_{n-4}, \quad n \geq 4$$

iv. Use (iii) to find I_5 .**2**v. Use the substitution $u = \frac{\pi}{4} - x$ to show that**2**

$$I_n = 2^n \int_0^{\frac{\pi}{4}} \left(\frac{1}{1 + \tan x} \right)^n dx$$

vi. Hence or otherwise evaluate

1

$$\int_0^{\frac{\pi}{4}} \left(\frac{1}{1 + \tan x} \right)^5 dx$$

End of paper.

$$1. \quad z = -1 - i = \sqrt{2} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right) = \sqrt{2} e^{\frac{5\pi i}{4}}$$

$$w = 1 + i\sqrt{3} = 2 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = 2 e^{i\pi/3}$$

$$\frac{z}{w} = \frac{\sqrt{2} e^{\frac{5\pi i}{4}}}{2 e^{i\pi/3}} = \frac{1}{\sqrt{2}} e^{i(\frac{5\pi}{4} - \frac{\pi}{3})} = \frac{1}{\sqrt{2}} e^{i\frac{11\pi}{12}}$$

(D)

$$2. \quad \vec{u} = 4\vec{i} + 4\vec{j} - 2\vec{k}$$

$$|\vec{u}|^2 = 16 + 16 + 4 = 36 \quad \therefore |\vec{u}| = 6$$

$$\hat{u} = \frac{1}{6} \begin{bmatrix} 4 \\ 4 \\ -2 \end{bmatrix}$$

$$\frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

(B)

$$3. \quad \int \frac{\cos^3 x - \sin^3 x}{\cos x - \sin x} dx = \int (\cos^2 x + \cos x \sin x + \sin^2 x) dx$$

$$= \int (1 + \cos x \sin x) dx = x + \frac{\sin^2 x}{2} + C$$

(C)

$$4. \quad V_T = \sqrt{\frac{g}{k}} = \sqrt{\frac{9.8}{\frac{1}{40}}} = \sqrt{4 \times 98} = 19.8 \text{ m s}^{-1}$$

(A)

$$5. \quad \left(1 + \frac{1}{w}\right)^{2020} \left(1 + \frac{1}{w^2}\right)^{2021}$$

$$= \left[\left(1 + \frac{1}{w}\right) \left(1 + \frac{1}{w^2}\right) \right]^{2020} \left(1 + \frac{1}{w^2}\right)$$

$$= \left[\frac{1+w}{w} \times \frac{1+w}{w^2} \right]^{2020} \left(\frac{1+w^2}{w^2} \right)$$

$$= \left[\frac{-w^2}{w} \times \frac{-w}{w^2} \right]^{2020} \left(\frac{-w}{w^2} \right)$$

$$= 1 \times \left(\frac{-1}{w} \right) = \frac{-1}{w}$$

(A)

$$1 + w + w^2 = 0$$

$$1 + w = -w^2$$

$$1 + w^2 = -w$$

$$6. \text{proj}_{\vec{OB}} \vec{OA} = \frac{(\vec{OA} \cdot \vec{OB})(\vec{OB})}{|\vec{OB}|^2}$$

$$\begin{aligned}\vec{OA} \cdot \vec{OB} &= 4 \times (-1) + 2 \times 1 + (-3) \times 1 \\ &= -4 + 2 - 3 \\ &= -5\end{aligned}$$

$$|\vec{OB}|^2 = (-1)^2 + 1^2 + 1^2 = 3$$

$$\text{proj}_{\vec{OB}} \vec{OA} = \frac{-5}{3} \vec{OB} = \frac{-5}{3} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \frac{5}{3} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} \quad (A)$$

$$7. |z| - z - 4(1 - 2i) = 0$$

$$z = |z| - 4 + 8i$$

$$|z|^2 = (|z| - 4)^2 + 64$$

$$|z|^2 = |z|^2 - 8|z| + 16 + 64$$

$$\therefore 8|z| = 80 \quad \therefore |z| = 10 \quad \therefore |z|^2 = 100 \quad (C)$$

$$8. \frac{v dv}{du} = -3v(1+v)$$

$$\frac{-1}{3} \frac{dv}{1+v} = du \quad \therefore \quad u = -\frac{1}{3} \int \frac{dv}{1+v} \quad (A)$$

$$\begin{aligned}(9). \frac{d}{dx} \int_x^{x^2} \frac{1}{t-1} dt &= \frac{1}{x^2-1} \times 2x - \frac{1}{x-1} \times 1 \\ &= \frac{2x^2 - 2x - x^2 + 1}{(x^2-1)(x-1)} = \frac{x^2 - 2x + 1}{(x^2-1)(x-1)} = \frac{(x-1)^2}{(x-1)^2(x+1)} = \frac{1}{x+1}\end{aligned}$$

10.

(C)

(D)

11.

$$(a) \quad z = 2 + 2i\sqrt{3} = 4e^{i\pi/3}$$

$$w = \sqrt{3} + i = 2e^{i\pi/6}$$

$$(i) \quad zw = 8e^{i\pi/2} = 8i$$

$$(ii) \quad \frac{z}{w} = \frac{4e^{i\pi/3}}{2e^{i\pi/6}} = 2e^{i\pi/6} = w$$

$$(b) \quad \int \frac{x}{\sqrt{1-x}} dx = \int \frac{x-1}{\sqrt{1-x}} dx + \int \frac{1}{\sqrt{1-x}} dx$$

$$= - \int \sqrt{1-x} dx + \int (1-x)^{-1/2} dx$$

$$= \frac{2}{3}(1-x)^{3/2} - 2(1-x)^{1/2} + C$$

$$= 2\sqrt{1-x} \left(\frac{1-x}{3} - 1 \right) + C$$

$$= 2\sqrt{1-x} \left(\frac{1-x-3}{3} \right) + C$$

$$= \frac{-2}{3}(x+2)\sqrt{1-x} + C$$

$$(c) \quad (i) \quad \sqrt{6i-8} = \sqrt{1^2-3^2+2 \times 1 \times 3i} = \sqrt{(1+3i)^2} = \pm(1+3i)$$

$$(ii) \quad 2z^2 - (3+i)z + 2 = 0$$

$$\Delta = [-(3+i)]^2 - 4 \times 2 \times 2$$

$$= 9+6i-1-16$$

$$= 6i-8$$

$$z = \frac{3+i \pm (1+3i)}{4} \begin{cases} 1+i \\ \frac{1-i}{2} \end{cases}$$

$$(d) \quad (i) \quad \frac{16}{(x^2+4)(2-x)} = \frac{ax+b}{x^2+4} + \frac{c}{2-x}$$

$$\begin{aligned} 16 &= (ax+b)(2-x) + c(x^2+4) \\ &= 2ax + 2b - ax^2 - bx + cx^2 + 4c \\ &= (c-a)x^2 + (2a-b)x + 2b + 4c \end{aligned}$$

$$c-a=0 \quad \therefore c=a$$

$$2a-b=0 \quad \therefore b=2a$$

$$2b+4c=16 \quad \therefore b+2c=8 \quad \therefore 2a+2a=8 \quad \therefore a=2$$

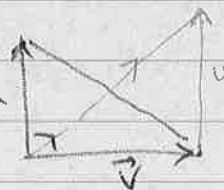
$$\therefore c=2 \quad \text{and} \quad b=4$$

$$\begin{aligned} (ii) \quad \int \frac{1}{(x^2+4)(2-x)} dx &= \frac{1}{16} \int \frac{16}{(x^2+4)(2-x)} dx \\ &= \frac{1}{16} \left[\int \frac{2x+4}{x^2+4} dx + 2 \int \frac{dx}{2-x} \right] \\ &= \frac{1}{16} \left[\int \frac{2x}{x^2+4} dx + \int \frac{4}{x^2+4} dx + 2 \int \frac{dx}{2-x} \right] \\ &= \frac{1}{16} \left[\ln(x^2+4) + 4 \times \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) - 2 \ln|2-x| \right] + C \\ &= \frac{1}{16} \ln(x^2+4) + \frac{1}{8} \tan^{-1}\left(\frac{x}{2}\right) - \frac{1}{8} \ln|2-x| + C \end{aligned}$$

$$(e) \quad \vec{u} = 2\vec{i} + 2\vec{j} - 3\vec{k} \quad \vec{v} = \vec{i} + 2\vec{j} + 2\vec{k}$$

$$\begin{aligned} (i) \quad \vec{u} \cdot \vec{v} &= 2(1) + 2(2) + (-3)(2) \\ &= 2 + 4 - 6 \\ &= 0 \quad \therefore \vec{u} \perp \vec{v} \end{aligned}$$

(ii) $\vec{u}$, $\vec{v}$ and $\vec{w}$ are the sides of right angle triangle, where $\vec{w}$ is the hypotenuse



$$\begin{aligned} \text{or } \vec{w} &= \vec{u} + \vec{v} \\ &= \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} \end{aligned}$$

$$\vec{u} + \vec{w} = \vec{v} \quad \text{or} \quad \vec{v} + \vec{w} = \vec{u}$$

$$\vec{w} = \vec{v} - \vec{u} \quad \text{or} \quad \vec{w} = \vec{u} - \vec{v}$$

$$\therefore \vec{w} = \pm \begin{pmatrix} 1 \\ 0 \\ -5 \end{pmatrix}$$

$$\text{Also } -(\vec{u} + \vec{v})$$

12.

If mn is odd then m and n are odd

(a) (i) contrapositive

if m or n are even, then mn is even.(ii) m is even and n is even then mn is even
(product of even is even).if m is even and n is odd

$$m = 2k \text{ and } n = 2p+1$$

$$mn = 2k(2p+1) = 2(2kp+k) = 2q \text{ is even.}$$

$$(b) \quad v^2 = -9x^2 + 18x + 27 = -9(x^2 - 2x - 3) = -9(x-3)(x+1)$$

$$(i) \quad \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{1}{2} (-9(2x) + 18) \\ = -9x + 9 \\ = -3^2(x-1)$$

(ii) centre $x=1$, amplitude = 2(iii) Maximum acceleration $\therefore v=0$
 $\therefore x=-1$ or $x=3$

$$\text{at } x=-1, \ddot{x} = 18$$

$$\text{at } x=3, \ddot{x} = -18.$$

$$(c) (i) \quad \vec{OA} - \vec{OC} + \vec{OB} - \vec{OC} + \vec{OD} - \vec{OC} + \vec{OE} - \vec{OC} = 0$$

$$\vec{OA} + \vec{OB} + \vec{OD} + \vec{OE} = 4\vec{OC}$$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 4\vec{OC}$$

$$\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = 4\vec{OC}$$

$$\vec{OC} = \frac{1}{4} \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \end{pmatrix} \therefore C \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right).$$

$$(ii) \quad \vec{CA} = \vec{OA} - \vec{OC} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

$$\vec{CB} = \vec{OB} - \vec{OC} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

$$|\vec{CA}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$|\vec{CB}| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$\vec{CA} \cdot \vec{CB} = \frac{1}{2} \times \left(-\frac{1}{2}\right) + \left(-\frac{1}{2}\right) \times \left(\frac{1}{2}\right) + \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) = -\frac{1}{4}$$

$$\cos \angle ACB = \frac{\vec{CA} \cdot \vec{CB}}{|\vec{CA}| |\vec{CB}|} = \frac{-\frac{1}{4}}{\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}} = -\frac{1}{3}$$

$$\begin{aligned} \angle ACB &= 109^\circ 28' \\ &= 109.47^\circ \\ &= 109.5^\circ \end{aligned}$$

$$(iii) \quad \vec{CE} = \vec{OE} - \vec{OC} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

$$\vec{AB} = \vec{OB} - \vec{OA} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{AD} = \vec{OD} - \vec{OA} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\vec{AB} \cdot \vec{CE} = -1 \times \frac{1}{2} + 1 \times \frac{1}{2} + 0 \times \frac{1}{2} = 0 \therefore \vec{AB} \perp \vec{CE}$$

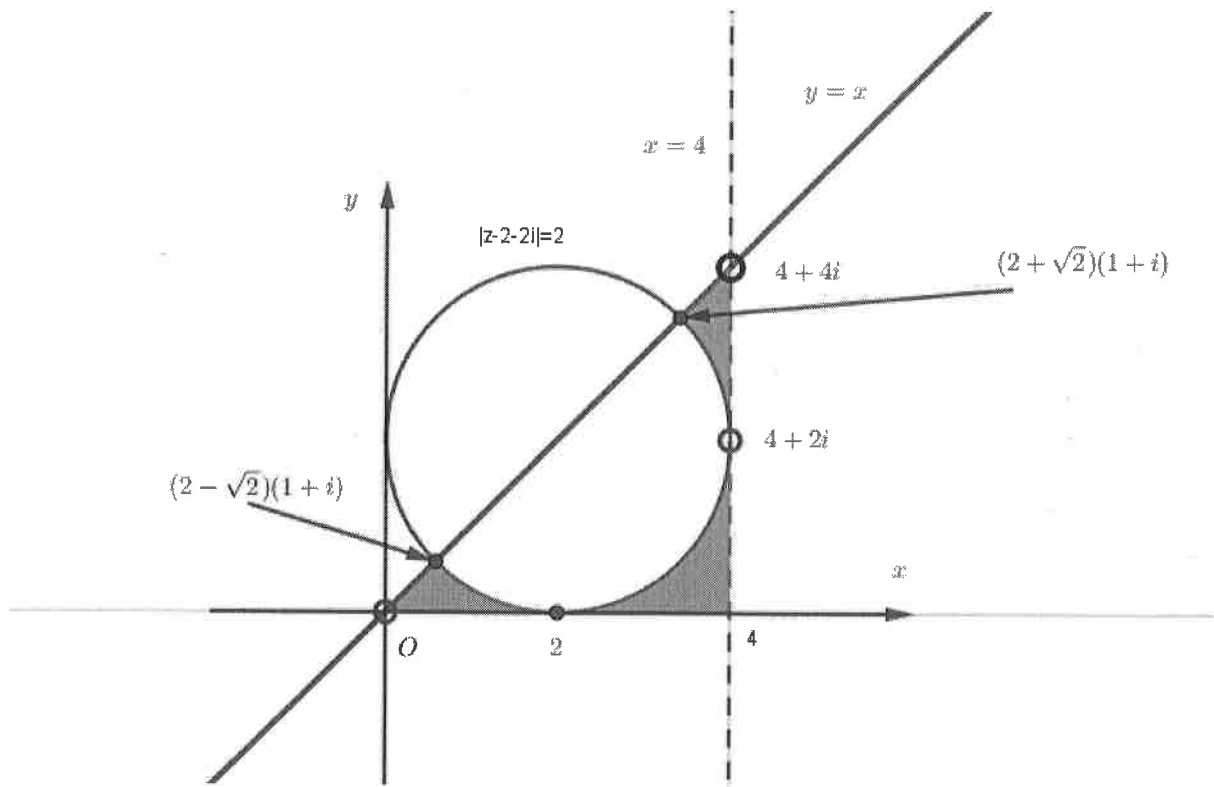
$$\vec{AD} \cdot \vec{CE} = -1 \times \frac{1}{2} + 0 \times \frac{1}{2} + 1 \times \frac{1}{2} = 0 \therefore \vec{AD} \perp \vec{CE}$$

$$\vec{CE} \perp \vec{AB} \text{ and } \vec{AD} \therefore \vec{CE} \parallel \text{line}$$

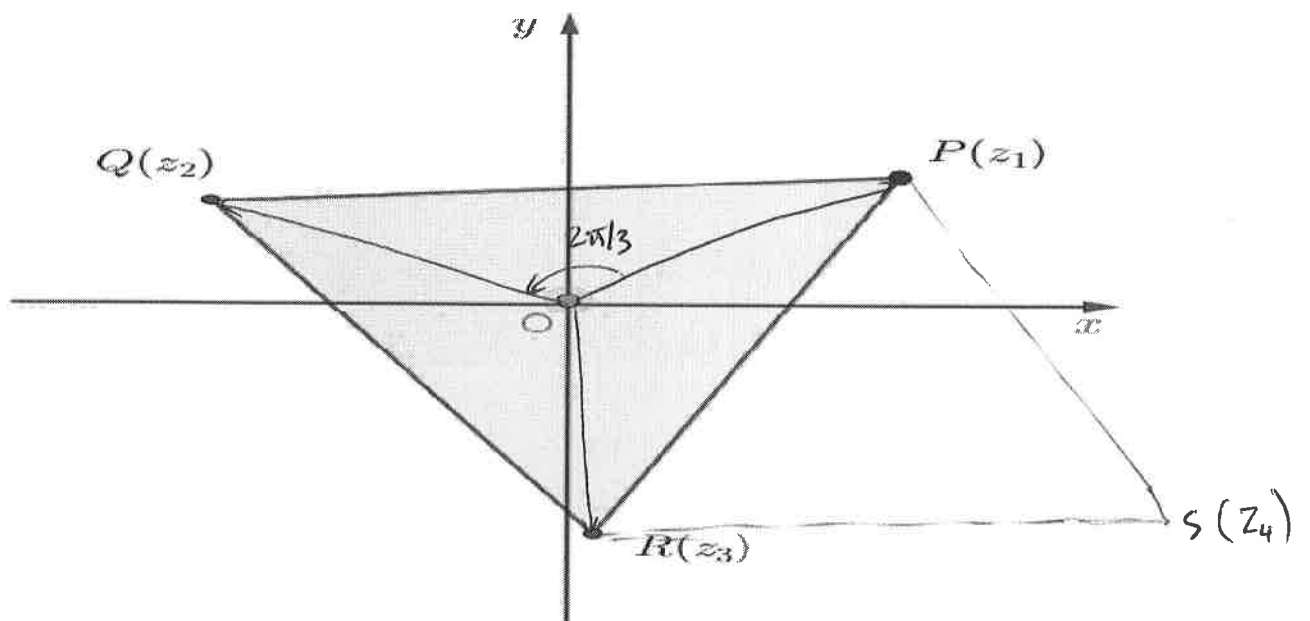
$$\text{Equation of line} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} + t \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} (1+t) \quad t \in \mathbb{R}.$$

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a.



b.



$$(i) \vec{OQ} = \vec{OP} \cos 2\pi/3 = \vec{OP} (e^{i\pi/3})^2 = \vec{OP} \omega^2$$

$$z_2 = z_1 \omega^2$$

Similarly, $\vec{OR} = \vec{OQ} \omega^2$

$$z_3 = z_1 \omega^2 \omega^2 = z_1 \omega^4$$

(ii) PQRS is a parallelogram $\therefore$

$$\vec{QP} = \vec{RS}$$

$$\vec{OP} - \vec{OQ} = \vec{OS} - \vec{OR}$$

$$\vec{OS} = \vec{OP} - \vec{OQ} + \vec{OR}$$

$$= z_1 - z_1 \omega^2 + z_1 \omega^4$$

$$= z_1 (1 - \omega^2 + \omega^4)$$

ω cube root of unity $\therefore \omega^3 = 1$ and $\omega^4 = \omega$

Also $1 + \omega + \omega^2 = 0 \therefore 1 + \omega = -\omega^2$

$$1 - \omega^2 + \omega^4 = -2\omega^2$$

$$z_4 = z_1 (-2\omega^2) = -2\omega^2 z_1$$

$$z_4 = (3 + 2\sqrt{3}) + (2 - 3\sqrt{3})i$$

$$(c) \quad (i) \quad \int_0^2 f(x) dx = \frac{1}{2} \pi (1)^2 = \pi/2$$

(area of semi circle with radius 1)

(1)

$$(ii) \quad \frac{2}{n} \left[f\left(\frac{2}{n}\right) + f\left(\frac{4}{n}\right) + \dots + f\left(\frac{2n-2}{n}\right) \right] - \frac{2}{n} f(1)$$

$$= \frac{2}{n} \sum_{k=1}^{n-1} \sqrt{\frac{2k}{n} \left(2 - \frac{2k}{n}\right)} - \frac{2}{n} f(1)$$

$$= \frac{2}{n} \sum_{k=1}^{n-1} \frac{2}{n} \sqrt{k(n-k)} - \frac{2}{n}$$

$$= 4 \sum_{k=1}^{n-1} \frac{\sqrt{k(n-k)}}{n^2} - \frac{2}{n}$$

(2)

$$(iii) \quad \frac{2}{n} \left[f\left(\frac{2}{n}\right) + f\left(\frac{4}{n}\right) + \dots + f\left(\frac{2n-2}{n}\right) \right] + \frac{2}{n} f(1)$$

$$= \frac{2}{n} \sum_{k=1}^{n-1} \sqrt{\frac{2k}{n} \left(2 - \frac{2k}{n}\right)} + \frac{2}{n} f(1)$$

$$= 4 \sum_{k=1}^{n-1} \frac{\sqrt{k(n-k)}}{n^2} + \frac{2}{n}$$

(2)

(iv) Area from (ii) $\leq$ Area of semi circle $\leq$ Area from (iii)

$$4 \sum_{k=1}^{n-1} \frac{\sqrt{k(n-k)}}{n^2} - \frac{2}{n} \leq \frac{1}{2} \pi \leq 4 \sum_{k=1}^{n-1} \frac{\sqrt{k(n-k)}}{n^2} + \frac{2}{n}$$

replace k by r (Note: As $n \rightarrow \infty$, $\frac{2}{n} \rightarrow 0$)

$$\therefore \lim_{n \rightarrow \infty} 4 \sum_{r=1}^n \frac{\sqrt{r(n-r)}}{n^2} \leq \frac{1}{2} \pi \leq \lim_{n \rightarrow \infty} 4 \sum_{r=1}^n \frac{\sqrt{r(n-r)}}{n^2}$$

$$\therefore 4 \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{r(n-r)}}{n^2} = \frac{1}{2} \pi$$

$$\therefore \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{r(n-r)}}{n^2} = \frac{1}{8} \pi.$$

(3)

Question 14:

$$\textcircled{a} \quad \ddot{x} = g - k v_A$$
$$t=0, \quad \dot{x}=0$$
$$x=0$$

$$\dot{v}_A = g - k v_A$$

$$\frac{k \dot{v}_A}{g - k v_A} = k$$

$$\int \frac{-k \dot{v}_A dt}{g - k v_A} = \int -k dt$$

$$\log(g - k v_A) = -k t + C_1$$

$$\text{At } t=0 \therefore C_1 = \log g$$

$$\log(g - k v_A) = -k t + \log g$$

$$\log\left(\frac{g - k v_A}{g}\right) = -k t \therefore t = -\frac{1}{k} \log\left(\frac{g - k v_A}{g}\right) \quad \textcircled{1}$$

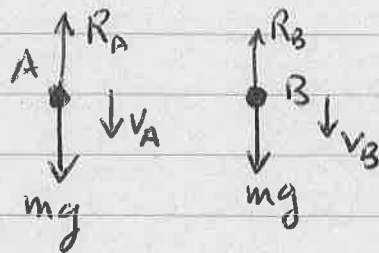
$$\ddot{x} = g - k v_B$$

$$t=0, \quad \dot{x} = u$$
$$x = 0$$

$$\frac{k \dot{v}_B}{g - k v_B} = k \therefore \int \frac{-k \dot{v}_B dt}{g - k v_B} = \int -k dt$$

$$\log(g - k v_B) = -k t + C_2$$

$$\text{At } t=0, \quad C_2 = \log(g - k u)$$



$$\log(g - kv_B) = -kt + \log(g - ku)$$

$$\begin{aligned} kt &= \log(g - ku) - \log(g - kv_B) \\ &= \log\left(\frac{g - ku}{g - kv_B}\right) \end{aligned}$$

$$t = \frac{1}{k} \log\left(\frac{g - ku}{g - kv_B}\right) \quad (2)$$

$$(1) - (2) \quad \therefore$$

$$\begin{aligned} T &= \frac{1}{k} \log\left(\frac{g - kv_A}{g}\right) - \frac{1}{k} \log\left(\frac{g - ku}{g - kv_B}\right) \\ &= \frac{1}{k} \log\left(\frac{g - ku}{g} \times \frac{g - kv_A}{g - kv_B}\right) \end{aligned}$$

$$(ii) \quad \ddot{x} = g - kv_B$$

$$v_B \frac{dv_B}{dn} = g - kv_B$$

$$\frac{v_B dv_B}{g - kv_B} = dn$$

$$\frac{-kv_B dv_B}{g - kv_B} = -k dn$$

$$\frac{(g - kv_B - g) dv_B}{g - kv_B} = -k dn$$

$$dv_B - \frac{g}{g - kv_B} dv_B = -k dn$$

$$V_B + \frac{g}{K} \ln(g - KV_B) = -Kx_B + C_3$$

$$t=0 \quad V_B = u, \quad x=0$$

$$C_3 = u + \frac{g}{K} \ln(g - Ku)$$

$$Kx = u + \frac{g}{K} \ln(g - Ku) - V_B - \frac{g}{K} \ln(g - KV_B)$$

$$= u - V_B + \frac{g}{K} \ln\left(\frac{g - Ku}{g - KV_B}\right)$$

$$x_B = \frac{1}{K} \left[u - V_B + \frac{g}{K} \ln\left(\frac{g - Ku}{g - KV_B}\right) \right]$$

(iii) use the result of (ii) for particle A

$$x_A = \frac{1}{K} \left[-V_A + \frac{g}{K} \ln\left(\frac{g}{g - KV_A}\right) \right]$$

$$\text{set } x_A = x_B$$

$$u - V_B + \frac{g}{K} \ln\left(\frac{g - Ku}{g - KV_B}\right) = -V_A + \frac{g}{K} \ln\left(\frac{g}{g - KV_A}\right)$$

$$\frac{g}{K} \ln\left(\frac{g - Ku}{g - KV_B}\right) - \frac{g}{K} \ln\left(\frac{g}{g - KV_A}\right) = V_B - V_A - u$$

$$\frac{g}{K} \ln\left(\frac{g - Ku}{g} \times \frac{g - KV_A}{g - KV_B}\right) = V_B - V_A - u$$

$$-\frac{1}{K} \ln\left(\frac{g - Ku}{g} \times \frac{g - KV_A}{g - KV_B}\right) = +\frac{u}{g} \quad \left(\text{at point of contact}\right)$$

$$T = \frac{u}{g} \quad (\text{using } T \text{ from (i)})$$

(b)

(i)

$$y_0 = \frac{1}{2} (\cos x + i \sin x)^0 + \frac{3}{2} (\cos x - i \sin x)^0$$

$$= \frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$$

$$y_1 = \frac{1}{2} (\cos x + i \sin x) + \frac{3}{2} (\cos x - i \sin x)$$

$$= 2 \cos x - i \sin x$$

Assume it is true for $n=k$ and $n=k-1$

$$y_k = \frac{1}{2} (\cos x + i \sin x)^k + \frac{3}{2} (\cos x - i \sin x)^k$$

$$= \frac{1}{2} (e^{ix})^k + \frac{3}{2} (e^{-ix})^k$$

$$y_k = \frac{1}{2} e^{ikx} + \frac{3}{2} e^{-ikx}$$

$$\text{and } y_{k-1} = \frac{1}{2} e^{i(k-1)x} + \frac{3}{2} e^{-i(k-1)x}$$

Prove it true for $n=k+1$

$$y_{k+1} = 2 \cos x y_k - y_{k-1}$$

$$= 2 \cos x \left(\frac{1}{2} e^{ikx} + \frac{3}{2} e^{-ikx} \right) - \frac{1}{2} e^{i(k-1)x} - \frac{3}{2} e^{-i(k-1)x}$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \therefore 2 \cos x = e^{ix} + e^{-ix}$$

~~$$y_{k+1} = \frac{1}{2} e^{i(k+1)x} + \frac{3}{2} e^{-i(k+1)x}$$~~

$$\begin{aligned}
 y_{k+1} &= (e^{ix} + e^{-ix}) \left(\frac{1}{2} e^{ikx} + \frac{3}{2} e^{-ikx} \right) - \frac{1}{2} e^{i(k-1)x} - \frac{3}{2} e^{-i(k-1)x} \\
 &= \frac{1}{2} e^{i(k+1)x} + \frac{1}{2} e^{i(k-1)x} + \frac{3}{2} e^{-i(k-1)x} + \frac{3}{2} e^{-i(k+1)x} - \frac{1}{2} e^{i(k-1)x} - \frac{3}{2} e^{-i(k-1)x} \\
 &= \frac{1}{2} e^{i(k+1)x} + \frac{3}{2} e^{-i(k+1)x}
 \end{aligned}$$

True for $n = k+1$

Hence by mathematical induction
it is true for all $n \geq 0$.

$$\begin{aligned}
 \text{(ii)} \quad \lim_{n \rightarrow \infty} \frac{y_{n+1}}{y_n} &= \lim_{n \rightarrow \infty} \frac{\frac{1}{2} e^{i(n+1)x} + \frac{3}{2} e^{-i(n+1)x}}{\frac{1}{2} e^{inx} + \frac{3}{2} e^{-inx}} \\
 &= \lim_{n \rightarrow \infty} \frac{e^{i(n+1)x} \left[\frac{1}{2} + \frac{3}{2} e^{-2i(n+1)x} \right]}{e^{inx} \left[\frac{1}{2} + \frac{3}{2} e^{-2inx} \right]} \\
 &= e^{ix} \lim_{n \rightarrow \infty} \frac{\frac{1}{2} + \frac{3}{2} e^{-2i(n+1)x}}{\frac{1}{2} + \frac{3}{2} e^{-2inx}} \\
 &= e^{ix}
 \end{aligned}$$

Question 15:

$$(a) (i) \quad S = 1 + 2\omega + 3\omega^2 + \dots + n\omega^{n-1}$$
$$wS = \omega + 2\omega^2 + \dots + (n-1)\omega^{n-1} + n\omega^n$$

$$S - wS = 1 + \omega + \omega^2 + \dots + \omega^{n-1} - n\omega^n$$
$$= -n\omega^n \quad \text{since } (1 + \omega + \dots + \omega^{n-1} = 0)$$

$$S(1-\omega) = -n\omega^n = -n \quad \text{since } \omega^n = 1$$

$$S = \frac{n}{\omega - 1}$$

$$(ii) \quad Z^n - 1 = (Z-1)(Z-\omega)(Z-\omega^2) \dots (Z-\omega^{n-1})$$

$$\frac{Z^n - 1}{Z - 1} = (Z-\omega)(Z-\omega^2) \dots (Z-\omega^{n-1})$$

$$1 + Z + Z^2 + \dots + Z^{n-1} = (Z-\omega)(Z-\omega^2) \dots (Z-\omega^{n-1})$$

$$\text{For } Z=1, \quad n = (1-\omega)(1-\omega^2) \dots (1-\omega^{n-1})$$

$$(iii) \quad p(Z) = (Z-\omega)(Z-\omega^2) \dots (Z-\omega^{n-1})$$

$$\frac{p'(Z)}{p(Z)} = \frac{1}{Z-\omega} + \frac{1}{Z-\omega^2} + \dots + \frac{1}{Z-\omega^{n-1}}$$

$$p(Z) = 1 + Z + Z^2 + \dots + Z^{n-2} + Z^{n-1}$$

$$p'(Z) = 1 + 2Z + 3Z^2 + \dots + (n-2)Z^{n-3} + (n-1)Z^{n-2}$$

$$p'(1) = 1 + 2 + 3 + \dots + n-1 = \frac{(n-1)n}{2}$$

$$p(1) = n$$

$$\frac{p'(1)}{p(1)} = \frac{\frac{(n-1)n}{2}}{n} = \frac{n-1}{2} \quad \therefore \frac{1}{1-\omega} + \frac{1}{1-\omega^2} + \dots + \frac{1}{1-\omega^{n-1}} = \frac{n-1}{2}$$

$$(b) \quad x = \frac{a}{a-b}, \quad y = \frac{b}{b-c}, \quad z = \frac{c}{c-a}$$

$$(i) \quad x-1 = \frac{a}{a-b} - 1 = \frac{a-a+b}{a-b} = \frac{b}{a-b}$$

$$y-1 = \frac{b}{b-c} - 1 = \frac{b-b+c}{b-c} = \frac{c}{b-c}$$

$$z-1 = \frac{c}{c-a} - 1 = \frac{c-c+a}{c-a} = \frac{a}{c-a}$$

$$\begin{aligned} (x-1)(y-1)(z-1) &= \frac{b}{a-b} \times \frac{c}{b-c} \times \frac{a}{c-a} \\ &= \frac{a}{a-b} \times \frac{b}{b-c} \times \frac{c}{c-a} \\ &= xyz \end{aligned}$$

$$(ii) \quad (x-1)(y-1)(z-1) = xyz$$

$$(x-1)(yz-y-z+1) = xyz$$

$$xyz - xy - xz + x - yz + y + z - 1 = xyz$$

$$x + y + z = xy + yz + zx + 1$$

$$(iii) \quad \left(\frac{2a-b}{a-b}\right)^2 + \left(\frac{2b-c}{b-c}\right)^2 + \left(\frac{2c-a}{c-a}\right)^2$$

$$= \left(1 + \frac{a}{a-b}\right)^2 + \left(1 + \frac{b}{b-c}\right)^2 + \left(1 + \frac{c}{c-a}\right)^2$$

$$= (1+x)^2 + (1+y)^2 + (1+z)^2$$

$$= 1 + 2x + x^2 + 1 + 2y + y^2 + 1 + 2z + z^2$$

$$= 3 + x^2 + y^2 + z^2 + 2(x+y+z)$$

$$= 3 + x^2 + y^2 + z^2 + 2(xy + yz + zx + 1)$$

$$= 5 + x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$= 5 + (x+y+z)^2 \geq 5.$$

Question 16 :

$$I_n = \int_0^{\pi/4} (1 + \tan x)^n dx$$

$$(1) \quad I_n = \int_0^{\pi/4} (1 + \tan x)^{n-2} (1 + \tan x)^2 dx$$

$$= \int_0^{\pi/4} (1 + \tan x)^{n-2} (1 + 2 \tan x + \tan^2 x) dx$$

$$= \int_0^{\pi/4} (1 + \tan x)^{n-2} (1 + \tan^2 x) dx + 2 \int_0^{\pi/4} \tan x (1 + \tan x)^{n-2} dx$$

$$= \int_0^{\pi/4} \frac{(1 + \tan x)^{n-1}}{n-1} dx + 2 \int_0^{\pi/4} (1 + \tan x - 1) (1 + \tan x)^{n-2} dx$$

$$= \frac{(1 + \tan \pi/4)^{n-1}}{n-1} - \frac{(1 + \tan 0)^{n-1}}{n-1} + 2 \int_0^{\pi/4} (1 + \tan x)^{n-1} dx - 2 \int_0^{\pi/4} (1 + \tan x)^{n-2} dx$$

$$= \frac{2^{n-1} - 1}{n-1} + 2 I_{n-1} - 2 I_{n-2} .$$

$$(ii) \quad I_0 = \int_0^{\pi/4} du = \pi/4$$

$$\begin{aligned} I_1 &= \int_0^{\pi/4} (1 + \tan u) du = \left[u - \ln \cos u \right]_0^{\pi/4} \\ &= \pi/4 - \ln \cos \pi/4 - 0 \\ &= \pi/4 - \ln \frac{1}{\sqrt{2}} \\ &= \pi/4 + \frac{\ln 2}{2} \end{aligned}$$

$$\begin{aligned} I_2 &= \frac{2^2 - 1}{1} + 2 I_1 - 2 I_0 \\ &= 1 + \pi/2 + \ln 2 - 2(\pi/4) \\ &= 1 + \ln 2 \end{aligned}$$

$$\begin{aligned} I_3 &= \frac{2^3 - 1}{2} + 2 I_2 - 2 I_1 \\ &= \frac{3}{2} + 2 + 2 \ln 2 - 2 \left(\pi/4 + \frac{\ln 2}{2} \right) \\ &= \frac{7}{2} + \ln 2 - \pi/2 \end{aligned}$$

$$\begin{aligned} I_4 &= \frac{2^4 - 1}{3} + 2 I_3 - 2 I_2 \\ &= \frac{7}{3} + 7 + 2 \ln 2 - \pi - 2 - 2 \ln 2 \\ &= \frac{22}{3} - \pi \end{aligned}$$

$$\begin{aligned} I_5 &= \frac{2^5 - 1}{4} + 2 I_4 - 2 I_3 \\ &= \frac{15}{4} + \frac{44}{3} - 2\pi - 7 - 2 \ln 2 + \pi \end{aligned}$$

$$\boxed{I_5 = \frac{137}{12} - 2 \ln 2 - \pi}$$

$$(iii) \quad I_n = \frac{2^{n-1} - 1}{n-1} + 2I_{n-1} - 2I_{n-2} \quad (1)$$

$$I_{n-1} = \frac{2^{n-2} - 1}{n-2} + 2I_{n-2} - 2I_{n-3} \quad (2)$$

$$I_{n-2} = \frac{2^{n-3} - 1}{n-3} + 2I_{n-3} - 2I_{n-4} \quad (3)$$

$$(2) + (3) :$$

$$I_{n-1} + I_{n-2} = \frac{2^{n-2} - 1}{n-2} + \frac{2^{n-3} - 1}{n-3} + 2I_{n-2} - 2I_{n-4}$$

$$\therefore I_{n-1} - I_{n-2} = \frac{2^{n-2} - 1}{n-2} + \frac{2^{n-3} - 1}{n-3} - 2I_{n-4}$$

$$(1) \therefore I_n = \frac{2^{n-1} - 1}{n-1} + 2(I_{n-1} - I_{n-2})$$

$$= \frac{2^{n-1} - 1}{n-1} + 2 \left(\frac{2^{n-2} - 1}{n-2} + \frac{2^{n-3} - 1}{n-3} - 2I_{n-4} \right)$$

$$\boxed{I_n = \frac{2^{n-1} - 1}{n-1} + 2 \cdot \frac{2^{n-2} - 1}{n-2} + 2 \cdot \frac{2^{n-3} - 1}{n-3} - 4I_{n-4}}$$

$$(iv) \quad I_5 = \frac{2^4 - 1}{4} + 2 \cdot \frac{2^3 - 1}{3} + 2 \cdot \frac{2^2 - 1}{2} - 4I_1$$

$$= \frac{15}{4} + \frac{14}{3} + 3 - 4 \left(\frac{\pi}{4} + \frac{1}{2} \ln 2 \right)$$

$$= \frac{137}{12} - 2 \ln 2 - \pi$$

$$(V) \quad I_n = \int_0^{\pi/4} (1 + \tan x)^n dx$$

$$= \int_{\pi/4}^0 \left(1 + \tan\left(\frac{\pi}{4} - u\right)\right)^n (-du)$$

$$= \int_0^{\pi/4} \left(1 + \tan\left(\frac{\pi}{4} - u\right)\right)^n du$$

$$= \int_0^{\pi/4} \left(1 + \frac{\tan \frac{\pi}{4} - \tan u}{1 + \tan \frac{\pi}{4} \tan u}\right)^n du$$

$$= \int_0^{\pi/4} \left(\frac{2}{1 + \tan u}\right)^n du$$

$$= 2^n \int_0^{\pi/4} \left(\frac{1}{1 + \tan u}\right)^n du$$

$$= 2^n \int_0^{\pi/4} \left(\frac{1}{1 + \tan x}\right)^n dx$$

$$(vi) \int_0^{\pi/4} \left(\frac{1}{1+\tan x} \right)^5 dx = \frac{I_5}{2^5}$$

$$= \frac{1}{32} \left(\frac{137}{12} - 2 \ln 2 - \pi \right).$$
